Yashveer Jain

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EDUCATION

Master of Engineering (Robotics) (GPA:3.85/4) Aug 2022–May 2024
University of Maryland – College Park (UMD), USA
Robot Perception, Becoming a Design Thinker, NLP, Visual Learning & Recognition, Deep Learning.

Bachelor of Engineering (Mechatronics) (GPA:9.44/10) July 2016- July 2020
Manipal University Jaipur, Rajasthan, India

WORK EXPERIENCE

Embedded AI Intern - Renesas Electronics, USA May 2023–Aug 2023

- *Trained and deployed SVM and logistic regression models on Renesas RA6 Series and RL78 microscontroller, achieving 95% accuracy for an Asset Tracking Project*, enabling efficient asset management.
- *Ensured success of edge computing projects* by developing and implementing machine learning models optimized for resource-constrained embedded systems.

Machine learning Engineer - AIMonk Labs, India Aug 2020–Aug 2022

- ***Developed and deployed a robust testing infrastructure for AI models***, integrating FastAPI backend on AWS EC2 instance, streamlining the model development lifecycle.
- ***Optimized OCR algorithms using Transformers implemented on PyTorch, achieving a 60% accuracy boost***, significantly improving text recognition capabilities.
- ***Enhanced car detection AI by 20% using DeepLabv3 and YOLOv7 integration***, leading to more accurate object detection in autonomous systems.
- ***Collaborated with cross-functional teams on Azure DevOps for model deployment on AWS SageMaker***, ensuring seamless integration and deployment of machine learning models.

Research Intern - Swaayatt Robotics, India Jan 2020–May 2020

- Implemented ***Kalman and particle filters*** with ***LK optical flow using OpenCV*** for vehicle and pedestrian tracking in autonomous driving systems ***increasing the detection accuracy by 20%.***

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Maryland (Fall 2022-Spring 2023) Aug 2022-May 2023
TinyDepth: Generalized Neural Metric Depth for Palm-sized Robots. Under Review Nature npj Robotics.
Designed sub-6cm microdrone with embedded Raspberry Pi and ROS-based vision pipeline for autonomous planning and control (Perception & Robotics Group, Prof. Yiannis Aloimonos).

PROJECTS

SVM-based Image Classification with Multithreading (C++)- GitHub Mar 2024-May 2024
Implemented a Support Vector Machine (SVM) for image classification in C++. Utilized multithreading for improved performance on the CIFAR-10 dataset.

Home Automation - GitHub Jun 2021-Aug 2021
Established a home server for device connectivity ***using FastAPI, MQTT, Arduino, Raspberry Pi***. Demonstrating proficiency in hardware integration and IoT systems.

Home Automation App- GitHub April 2021-Aug 2021
Developed a home automation app using ***Flutter, Dart, Firebase, and SQLite*** with real-time database, dynamic UI, and device management for controlling electrical appliances. Enhanced with room/device management, status updates, and time-based greetings.

UnDeepVO (Unsupervised Deep Learning Visual Odometry) - GitHub Jan 2020 - May 2020
Implemented unsupervised deep learning visual odometry paper using python and tensorflow, to handle depth estimation and transformation (translation and rotation of camera) using monocular camera images.

SKILLS AND LANGUAGES

Machine Learning: TensorFlow, PyTorch, Pandas, Matplotlib, OpenCV, Scikit-learn, Huggingface, TensorRT.
Programming Languages: Python, C, C++, Cuda, Flutter, Dart, sqlite
Development Tools: Git, Docker, Agile Development
Cloud Platforms: AWS (Certified Cloud Practitioner), Azure
Robotics: ROS (Robot Operating System), ROS2, Arduino, Raspberry Pi, Nvidia Jetson.

CERTIFICATE

Building Transformer-Based Natural Language Processing Applications (Nvidia) Feb 2024
Training and Deploying **BERT Transformer Model** on **NER and Text classification** and deployed on **Triton Server**.

CUDA programming Masterclass with C++ (Udemy) Mar 2024
Gained knowledge in implementing CUDA algorithms and managing memory, including shared memory and global memory, as well as handling synchronization using streams.